Phase 1: AI Project Proposal
Optimizing Campus Course Selection and Scheduling

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Project topic: Students struggle to schedule their timetables well through AIMS.
Target users/stakeholders: Students using AIMS, particularly students with large flexibility in course scheduling.

1 Problem Definition & Scope

1.1 Problem

Scheduling a well organised timetable with one's desired lessons is difficult, especially when allowed to choose from a large list of courses with many timeslots each.

Current situation: The AIMS platform is ill suited to effective scheduling as its interface makes considering possible timetables difficult, let alone deciding on the best of one's possible timetables.

Affected users: Students with large flexibility in selecting courses, particularly students with a large number of GE courses to select from or Advanced Standing students with too many leftover credits.

Importance: Simplifying the course selection process would allow students to complete the Add/Drop process more efficiently. This could also improve the performance of the AIMS website by reducing the duration of the initial rush when students swarm to select their preferred courses.

1.2 Scope

Generate a list of valid schedules by brute force based on the user's selection of courses and rank them to prefer features of schedules based on selected criteria. Selection criteria would include selecting for lessons that avoid certain hours or days, or lessons that are early or late in the day Return the top scoring schedules from best to worst for the user to select from.

Features excluded from scope: The ability for the AI to select the actual courses being taken alongside scheduling them, including ranking the suitability of courses by topic, difficulty, type, and location (Could be suitable for a more advanced implementation but beyond the scope of this version). Producing a strong schedule by AI directly (Producing all valid schedules by brute force and selecting from them is a simpler task).

2 AI Task Formulation

Task: Rank a set of timetables based on selected selection criteria, including, but not necessarily limited to, time of day, day of week, and consecutiveness

Inputs: List of occupied schedule slots / Slot in schedule's status as occupied / unoccupied (6 days of the week × 13 possible hours per day = 78 binary inputs)

Outputs: Scores based on various criteria (eg: "earliness", "lateness", "consecutiveness", etc; one output per criteria, potentially with an additional final layer for an overall aggregate score)

Constraints: Scores outputted from 0-100. Model will only be fed valid timetables (ie. without collisions)

Evaluation: Judge accuracy of suggestions; Scores should agree with the opinion of users.

Data source: Survey students for their opinion on how a set of generated timetables should be scored. No privacy, consent, or access considerations needed.

3 Proposed AI Approaches

Present multiple plausible AI approaches. For each approach, briefly explain the core method, required data/resources, expected workflow, and practical feasibility. Divergent solutions are encouraged; do not present only one method under different names.

3.1 Convolutional NN

- Benefits
  - Better at preserving adjacency data
    - More accurate scores regarding "adjacency"
- Weaknesses
  - (to be written)
- Core method
  - Run a convolutional kernel across the squares of the timetable before feeding the results through a simple fully connected neural network.
- Required resources
  - A laptop running Pytorch
- Required data
  - A set of randomly generated schedules and scores assigned by volunteers.

3.2 Linear regression

- Benefits
  - Trivial to optimise.
- Weaknesses
  - (to be written)
- Core method
  - (to be written)
- Required resources
  - (to be written)
- Required data
  - (to be written)

RLHF?

Table 1: Comparison of candidate AI approaches.

Approach	Expected benefit	Main requirement	Main limitation
CNN	Learns timetable patterns	Labeled schedule ratings	Needs more data
Reinforement Learning	Base on user's preferrence	The feedback for the result	Design the feedback model

4 Comparative Analysis

A compact comparison table is encouraged when it saves space.

5 Expected Impact & Project Feasibility

Expected benefit: Easier scheduling, more efficient time management

Novelty: Likely very novel?

Value: High, helps students and the university schedule more efficiently, balance the preferences of students and professors

Data: Data of courses (scraped from AIMS Master Timetable), user preference data (for which timetables are scored higher)

Data collection plan: Survey students to select preferred timetables from generated timetables or score generated timetables. (eg. Ask students to score a set of 5 timetables from 0-100 based on the set of criteria listed above.)

Implementation: Create a dataset with scored timetables and optimise the regression/convolutional model to assign appropriate scores to schedules.

Tools: uhhh

Risks: Extremely minor: No identifying information needs to be collected (eg. name, class, major) and all collected information is not

Mitigation: N/A

Access to hardware: For Linear regression: Trivial (one laptop)

Access to software:

Access to data:

References

[1] Lance Le, "A Sample Title," *A sample journal*, 2023.

[2] Author Maple, "Rubik's Cube Methods," *Journal of Cube*, 2021.